Pongnapat Arayangkul

Robotics and Automation Engineering
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PROFILE

Robotics and Automation Engineering student with hands-on experience in **robot mechanism design, embedded systems, and robotic system integration**. Experienced in developing bio-inspired robotic mechanisms, custom embedded electronics, and experimental robotic platforms through research and engineering projects. Proficient in SOLIDWORKS, STM32 (C/C++), MATLAB/Simulink, and rapid prototyping. Passionate about translating research concepts into functional robotic systems through system-level design, experimental validation, and iterative engineering.

EDUCATION

King Mongkut's University of Technology Thonburi (KMUTT)

Aug 2023 – Expected 2027

Bachelor of Engineering in Robotics and Automation Engineering — Bangkok, Thailand

- GPA: 3.66 | Cumulative 3.47
- Scholarship: Teaching Assistant Scholarship (FRA232: Robot Structure Design, 2025)
- Relevant Coursework: Physical AI, Embedded Systems, Sensor & Actuator Development, Deep Reinforcement Learning

SKILLS

Languages: English (Advanced, IELTS 7.5, TOEIC 930/990), Thai (Native)
Mechanical Design: CAD/CAE (SOLIDWORKS, Fusion 360), FEA, Prototyping & Manufacturing (3D Printing, Machining)
Embedded Systems: STM32, Serial Communication Protocols, PCB Design & Fabrication, Power Budgeting
Programming: C/C++, Embedded C, Python, OOP, Custom Library Development, Git/GitHub, LaTeX
Robotics & Control: ROS 2 (Fundamentals), MATLAB/Simulink, Kinematic & Dynamic Modeling, Reinforcement Learning
Research & Engineering: Literature Review, Experimental Design, System Integration, Performance Evaluation
Soft Skills: Teamwork, Leadership, Technical Presentation, Requirement Analysis, Feasibility Study, Critical & Design Thinking

WORK EXPERIENCE

Research Intern – Diving Beetle-Inspired Amphibious Robot

Feb 2026 – Present

Bio-Inspired Robotics & Neural Engineering Lab, VISTEC — Rayong, Thailand

- Investigated **bio-inspired locomotion** for amphibious robots, focusing on terrestrial walking and **passive asymmetric force generation** for underwater swimming through morphological computation.
- Contributed to research and authored a **scientific manuscript** submitted to an international robotics conference.
- Designed a **cable-driven differential transmission** enabling **2-DOF actuation** while reducing distal leg inertia.
- Designed and integrated the **overall robot architecture**, including mechanical structures and electronics enclosures.
- Developed and validated **waterproof motor modules** and connector interfaces for underwater operation.

Teaching Assistant – SOLIDWORKS, Robot Structure Design, Scientific & Engineering Process

Aug 2024 – May 2026

Institute of Field Robotics (FIBO), KMUTT — Bangkok, Thailand

- Mentored **~75 students** through **7 technical workshops** on SOLIDWORKS and engineering principles, providing lab guidance, assignment support, grading, and technical consultation.
- Advised student teams on requirement analysis, system design and integration of a **1-DOF pick-and-place robot**.

Research Assistant – Solar Cell Cleaning Robot

Jun 2025 – Nov 2025

Institute of Field Robotics (FIBO), KMUTT — Bangkok, Thailand

- Optimized the brush mechanism through two design iterations and supported transporter and cleaner subsystem design and maintenance, simplifying the **control system** and improving **operational reliability**.
- Contributed to research under the **Electricity Generating Authority of Thailand (EGAT)**, developing robotics solutions for large-scale floating solar farms.

Vice President & Head of Mechanical Team – FIBOX Club

Nov 2023 – Jun 2025

Institute of Field Robotics (FIBO), KMUTT

- Led a **12-member mechanical team** across two ABU Robocon competition cycles, overseeing CAD design, prototyping, testing, and procurement of complex subsystems (grippers, flywheel launcher, omni/mecanum drives), while mentoring junior members.
- Drove the mechanical team to a **Top 24 national finish** and coordinated club operations and events, demonstrating leadership, resource management, and cross-functional teamwork.

PROJECTS

CBF-RL with Non-Ideal Cart-Pole Dynamics – Deep Reinforcement Learning Course Project
Apr 2026 – May 2026

Reinforcement Learning & Simulation

- Developed and evaluated safe reinforcement learning strategies by integrating **Control Barrier Functions (CBFs)** with **PPO** in **Isaac Lab**.
- Modeled realistic hardware non-idealities including motor saturation, voltage sag, gear backlash, and track constraints to improve **sim-to-real fidelity**.
- Analyzed deployment robustness using episode length, safety violation rate, hardware stress, and task performance across multiple training conditions.

Electroosmotic Soft Actuator – Sensor & Actuator Development Course Project
Jan 2026 – May 2026

Soft Actuator Design & Characterization

- Designed and developed an electroosmotic soft actuator using **electroosmotic flow** to generate actuation without conventional mechanical transmission.
- Redesigned the actuator from a rigid piston to a **compliant membrane architecture**, overcoming pressure generation and sealing limitations.
- Conducted experimental characterization of flow, pressure, displacement, and force, achieving up to **44.2 mm displacement** and **0.57 N output force**.

Embodied Hexapod Robot – Physical AI Course Project
Aug 2025 – May 2026

Hardware Engineer: Sensor Design & Embedded Electronics

- Designed and analyzed a **3-axis foot-force sensing module** using load cells to measure ground reaction forces for stable locomotion.
- Performed **Finite Element Analysis (FEA)** and redesigned the sensor mount into a compact, fully integrated final link to improve structural integrity and measurement accuracy.
- Designed the robot's **central electrical system**, including custom PCBs for battery monitoring, emergency switch handling, and system status indication, while selecting battery specifications through **power budgeting** to meet voltage, current, and runtime requirements.

RFID Access System with EEPROM Database and UI – Embedded Systems Course Project
Nov 2025 – Jan 2026

Embedded Firmware & System Integration

- Designed an RFID access control system using **STM32** and **MFRC522**, implementing UID authentication and a non-volatile EEPROM database with add/remove/verify functionality through an LCD interface.
- Implemented a **modular, non-blocking firmware architecture** using a **Finite State Machine**, integrating RFID, EEPROM, LCD, RTC, LED heartbeat, and buzzer feedback.
- Designed and fabricated a **custom PCB** integrating all system components; the system was **requested for future deployment in faculty facilities**.

Dynamic Modeling & Simulation of a 2-R Planar Pendulum – Kinematics Course Project
Oct 2025 – Nov 2025

Dynamics Modeling & Control Simulation

- Developed a complete **kinematic and dynamic model** of a 2-DOF planar serial manipulator using **DH parameters** and **Lagrangian formulation**, covering forward kinematics, inverse kinematics, and differential kinematics.
- Implemented a **MATLAB/Simulink simulation framework** integrating trajectory generation, inverse dynamics, and joint-space position/velocity control with gravity compensation.
- Validated analytical models against **Simscape Multibody** and evaluated **disturbance rejection**, demonstrating accurate end-effector recovery under external torque disturbances.

2-DOF Vertical RP Plotter – Industrial Automation Course Project
Jan 2025 – Jun 2025

Project Lead: Mechanical Design & Control Systems

- Led the project team in defining system requirements and applying requirement-driven design and system integration for a revolute–prismatic plotter.
- Designed mechanical components in **SOLIDWORKS**, performed mechanical calculations, coordinated with manufacturers for non-standard parts, and adapted the mechanical layout for electrical wiring and control box integration.
- Implemented PID tuning, **trajectory planning, and text/image plotting functions**, achieving **0.2 mm linear accuracy**, 0.5° rotational accuracy, and 600 mm/s linear velocity.

Swerve Drive Power Transmission – ABU Robocon 2025
Nov 2024 – Mar 2025

Design Engineer: Drivetrain Specialist

- Designed and optimized a **compact swerve drive** power transmission utilizing bevel gears, belts, and specialized clamping techniques for robust performance in a high-load environment.
- Integrated incremental encoder with Z-index homing for precise control.
- Identified traction and load issues during testing, highlighting the importance of realistic **testing environments**.